

Reflection on Technical Learnings

Battery Management System for Electric Scooter | ECE 445 Team 35

The biggest technical lesson I learned from the full BMS project is that battery management is a system-level problem, not just a sensing problem. At first, it is easy to think of a BMS as a collection of measurements: cell voltage, temperature, current, SOC, and fault flags. In practice, the value of the system comes from how those measurements are acquired, validated, interpreted, and acted upon under real operating conditions.

Current sensing was a major example of this. The project moved from relying mainly on open-circuit-voltage lookup toward coulomb counting with a shunt resistor and INA240 current-sense amplifier. This change made the design more realistic because an electric scooter is usually under dynamic load, where OCV is not always available or accurate. Designing the shunt path required thinking through signal scaling. With a 1 mOhm shunt and 100 V/V gain, 30 A maps to 3.0 V at the ADC, which uses most of the STM32 ADC range without saturation. The 12-bit ADC resolution corresponds to roughly 8 mA per count after gain and shunt conversion. I learned that accuracy is not just a sensor datasheet number; it depends on gain, ADC range, offset, drift, noise rejection, and how firmware integrates the result over time.

I also learned the importance of communication robustness. The LTC6811 data is safety-critical, so the isoSPI link needed noise immunity and error checking. Using differential, transformer-isolated communication helped make the master-slave connection more reliable in a noisy battery environment. On the viewer side, telemetry needed start-of-frame bytes, payload length, and CRC validation so corrupted packets would not be displayed as real pack data. This changed my view of firmware from simply transmitting values to enforcing trust in the data path.

Finally, the project taught me that verification should be quantitative. It was not enough to say that the board worked; we needed results such as 6.2 mV maximum voltage error, 1.8 C temperature error, 0 CRC errors over 500 isoSPI cycles, and fault assertion within 20 ms. These numbers made the design defensible. The main learning was that good engineering is the connection between design intent, physical implementation, and measured evidence. I also learned that clear telemetry makes debugging much faster, because it turns embedded behavior into observable engineering data.